

Dynamic IQC Duality

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I. PRELIMINARIES

A. Notation

$L_2^n[a, b]$ denotes the set of $\mathbb{R}^n$ -valued, Lebesgue square-integrable equivalence classes of functions on the spatial domain $[a, b] \subset \mathbb{R}$. The space $\mathbb{R}L_2^n[a, b]$ denotes the Cartesian product $\mathbb{R}^{n_1} \times L_2^{n_2}[a, b]$, where $\mathbf{n} = \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$, with inner product

$$\left\langle \begin{bmatrix} x_1 \\ \mathbf{x}_2 \end{bmatrix}, \begin{bmatrix} y_1 \\ \mathbf{y}_2 \end{bmatrix} \right\rangle_{\mathbb{R}L_2} := x_1^\top y_1 + \langle \mathbf{x}_2, \mathbf{y}_2 \rangle_{L_2}.$$

When clear from context, we occasionally omit the subscript and simply write $\langle \cdot, \cdot \rangle := \langle \cdot, \cdot \rangle_{\mathbb{R}L_2}$. Since the only spatial domain considered here is $[a, b]$, we typically omit the domain and write L_2^n or $\mathbb{R}L_2^n$, further omitting dimensions when clear from context. For a normed space X and Banach space Y , $\mathcal{L}(X, Y)$ denotes the Banach space of bounded linear operators from X to Y with induced norm

$$\|\mathcal{P}\|_{\mathcal{L}(X, Y)} = \sup_{\|\mathbf{x}\|=1} \|\mathcal{P}\mathbf{x}\|_Y,$$

and $\mathcal{L}(X) := \mathcal{L}(X, X)$. Our notational convention is to write functions in **bold** (e.g., $\mathbf{x}$, indicating $\mathbf{x} \in \mathbb{R}L_2$) and operators in calligraphic font (e.g., $\mathcal{P} \in \mathcal{L}(\mathbb{R}L_2)$). For a Hilbert space X and operator $\mathcal{A} \in \mathcal{L}(X)$, the adjoint $\mathcal{A}^*$ is defined by

$$\langle \mathbf{x}, \mathcal{A}\mathbf{y} \rangle_X = \langle \mathcal{A}^*\mathbf{x}, \mathbf{y} \rangle_X, \quad \forall \mathbf{x}, \mathbf{y} \in X.$$

The space $L_2^n[0, \infty)$ denotes $\mathbb{R}^n$ -valued square-integrable signals defined on the temporal domain $[0, \infty)$, so that $x(t) \in \mathbb{R}^n$. Similarly, $\mathbb{R}L_2^n[0, \infty)$ denotes $\mathbb{R}L_2$ -valued square-integrable temporal signals. For suitably differentiable $\mathbf{x} \in \mathbb{R}L_2^n[0, \infty)$, we write $\dot{\mathbf{x}} := \frac{\partial \mathbf{x}}{\partial t}$. For a Hilbert space $\mathbf{H}$, we denote by $\mathbf{L}_{\mathbf{H}}$ the space of square-integrable functions $u : [0, \infty) \rightarrow \mathbf{H}$ with inner product

$$\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{L}_{\mathbf{H}}} = \int_0^\infty \langle \mathbf{u}(t), \mathbf{v}(t) \rangle_{\mathbf{H}} dt.$$

We define the extended space $\mathbf{L}_{e, \mathbf{H}}$ to consist of functions $u : [0, \infty) \rightarrow \mathbf{H}$ such that

$$\int_0^T \|\mathbf{u}(t)\|_{\mathbf{H}}^2 dt < \infty \quad \forall T \geq 0.$$

We use $\mathbf{L}_{e, [a, b]}$ and $\mathbf{L}_{[a, b]}$ to denote $\mathbf{L}_{e, \mathbf{H}}$ and $\mathbf{L}_{\mathbf{H}}$ with $\mathbf{H} = \mathbb{R}L_2[a, b]$. For any Hilbert space $\mathbf{H}$, the truncation operator $p_\tau : \mathbf{L}_{e, \mathbf{H}} \rightarrow \mathbf{L}_{e, \mathbf{H}}$ is defined for $\mathbf{y} \in \mathbf{L}_{e, \mathbf{H}}$ by

$$(p_\tau \mathbf{y})(t) = \begin{cases} \mathbf{y}(t), & 0 \leq t \leq \tau, \\ 0, & \text{otherwise.} \end{cases}$$

Definition 1. Let $\mathbf{H}$ and $\mathbf{G}$ be Hilbert spaces. For all $\tau \geq 0$, an operator $G : \mathbf{L}_{\mathbf{H}} \rightarrow \mathbf{L}_{\mathbf{G}}$ is:

- **Causal** if $p_\tau G = p_\tau G p_\tau$.
- **Bounded on $\mathbf{L}_{\mathbf{H}}$** if $\|G\mathbf{v}\|_{\mathbf{L}_{\mathbf{G}}} \leq C\|\mathbf{v}\|_{\mathbf{L}_{\mathbf{H}}}$ for some $C \geq 0$ and all $\mathbf{v} \in \mathbf{L}_{\mathbf{H}}$.
- **Bounded on $\mathbf{L}_{e, \mathbf{H}}$** if $\|p_\tau G\mathbf{v}\|_{\mathbf{L}_{\mathbf{G}}} \leq C\|p_\tau \mathbf{v}\|_{\mathbf{L}_{\mathbf{H}}}$ for some $C \geq 0$ and all $\mathbf{v} \in \mathbf{L}_{e, \mathbf{H}}$.

Finally, for a given $\mathcal{K} \in \mathcal{L}(\mathbb{R}L_2^n, \mathbb{R}L_2^m)$, we define the operator $\mathcal{M}_{\mathcal{K}} : \mathbf{L}_{e, [a, b]}^{\mathbf{n}} \rightarrow \mathbf{L}_{e, [a, b]}^{\mathbf{m}}$ to compactly represent a multiplication operator. For $\mathbf{u} = [u_1 \ \cdots \ u_N]^\top$ with $\mathbf{u} \in \mathbb{R}L_2$, its action at time t is given by the block structure of $\mathcal{K}_{ij}$:

$$(\mathcal{M}_{\mathcal{K}}\mathbf{u})(t) := \begin{bmatrix} \mathcal{K}_{11} & \cdots & \mathcal{K}_{1n} \\ \vdots & \ddots & \vdots \\ \mathcal{K}_{m1} & \cdots & \mathcal{K}_{mn} \end{bmatrix} \begin{bmatrix} \mathbf{u}_1 \\ \vdots \\ \mathbf{u}_n \end{bmatrix} (t). \quad (1)$$

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B. PI operators

Partial Integral operators, also known as PI operators, are a class of bounded linear integral operators that are defined jointly on a vector space and Hilbert space.

Definition 2. We say $\mathcal{P} \in \Pi_4^{\mathbf{m} \times \mathbf{n}} \subset \mathcal{L}(\mathbb{R}L_2^{\mathbf{n}}, \mathbb{R}L_2^{\mathbf{m}})$, denoted by $\Pi \begin{bmatrix} P & Q_1 \\ Q_2 & \{R\} \end{bmatrix}$, if there exist a matrix P and matrix polynomials Q_1, Q_2, R_0, R_1 , and R_2 (of compatible dimension) such that,

$$\begin{aligned} \left[\left(\Pi \begin{bmatrix} P & Q_1 \\ Q_2 & \{R\} \end{bmatrix} \right) \begin{bmatrix} x \\ \mathbf{x} \end{bmatrix} \right] (s) &:= \begin{bmatrix} Px + \int_a^b Q_1(s) \mathbf{x}(s) ds \\ Q_2(s)x + \mathcal{R}\mathbf{x}(s) \end{bmatrix}, \\ (\mathcal{R}\mathbf{x})(s) &= R_0(s)\mathbf{x}(s) + \int_a^s R_1(s, \theta)\mathbf{x}(\theta) d\theta + \int_s^b R_2(s, \theta)\mathbf{x}(\theta) d\theta. \end{aligned}$$

The vector space of PI operators, given compatible dimensions, are closed under composition, addition, adjoint and concatenation – implying that Π_4 is a composition algebra [2].

C. Linear PI Inequalities (LPIs)

Definition 3. (LPIs) For given PI operators $\{\mathcal{E}_{i,j}, \mathcal{F}_{i,j}, \mathcal{G}_i\}$ and a convex linear functional $\mathcal{L}(\cdot)$, a linear PI inequality is a convex optimization of the following form

$$\begin{aligned} \min_{P_i, Q_{1i}, Q_{2i}, R_{0i}, R_{1i}, R_{2i}} & \mathcal{L}(\{P_i, Q_{1i}, Q_{2i}, R_{0i}, R_{1i}, R_{2i}\}) \\ \text{s.t.} & \sum_{j=1}^K \mathcal{E}_{ij}^* \Pi \begin{bmatrix} P_i & Q_{1i} \\ Q_{2i} & \{R_i\} \end{bmatrix} \mathcal{F}_{ij} + \mathcal{G}_i \succeq 0 \end{aligned} \quad (2)$$

Only LMIs are required to solve an optimization problem involving LPIs.

D. PIETOOLS

PIETOOLS is a MATLAB toolbox used for declaration, manipulation and solving LPIs [1]. For learning how to use PIETOOLS and its functionalities visit PIETOOLS homepage.

II. IQC FOR PIES

This section develops a comprehensive framework for analyzing stability and performance of interconnected systems in a Hilbert space, utilizing IQCs as the primary analytical tool. We begin by establishing the mathematical foundation for system interconnections and formalizing the key concepts of stability and performance. The IQC methodology is then systematically developed to characterize system uncertainties and derive rigorous conditions that guarantee robust stability and performance. A central contribution of this work lies in bridging IQC theory with μ -analysis for infinite dimensional systems, creating a framework for quantifying their robust stability and performance when those systems are subjected to structured uncertainties.

A. Description of uncertain systems in a Hilbert-space

We start by denoting the uncertainty operators as Δ that belongs to a structured set $\mathbf{\Delta}$ which satisfies the following fundamental property:

Assumption 1. $\Delta \in \mathbf{\Delta}$ implies that $\tau\Delta \in \mathbf{\Delta}$ for all $\tau \in [0, 1]$.

That is, the uncertainty set is closed under scaling by factors in $[0, 1]$, which is the standard homotopy assumption used in IQC theory. We consider operators that map external inputs to external outputs through a feedback interconnection between a linear (infinite dimensional) operator G and Δ . This interconnection structure is formally defined as follows

Definition 4. Given operators $G_\Delta : \mathbf{L}_{e,[a,b]}^{\mathbf{n}_w\Delta} \rightarrow \mathbf{L}_{e,[a,b]}^{\mathbf{n}_z\Delta}$, $G_{\Delta w} : \mathbf{L}_{e,[a,b]}^{\mathbf{n}_w} \rightarrow \mathbf{L}_{e,[a,b]}^{\mathbf{n}_z\Delta}$, $G_{z\Delta} : \mathbf{L}_{e,[a,b]}^{\mathbf{n}_w\Delta} \rightarrow \mathbf{L}_{e,[a,b]}^{\mathbf{n}_z}$, $G_{zw} : \mathbf{L}_{e,[a,b]}^{\mathbf{n}_w} \rightarrow \mathbf{L}_{e,[a,b]}^{\mathbf{n}_z}$, and $\Delta : \mathbf{L}_{e,[a,b]}^{\mathbf{n}_z\Delta} \rightarrow \mathbf{L}_{e,[a,b]}^{\mathbf{n}_w\Delta}$, we define an **interconnected system** $z = [G, \Delta](w)$, for inputs $w \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_w}$ and outputs $z \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_z}$, such that the signals $z_\Delta \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_z\Delta}$, $w_\Delta \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_w\Delta}$ satisfy the following relations

$$\begin{aligned} z_\Delta &= G_\Delta w_\Delta + G_{\Delta w} w, \\ z &= G_{z\Delta} w_\Delta + G_{zw} w, \\ w_\Delta &= \Delta z_\Delta. \end{aligned} \quad (3)$$

The interconnection structure described above captures the feedback interaction between the nominal system G and the uncertainty Δ , where w and z represent external inputs and performance outputs, respectively, while w_Δ and z_Δ form the internal feedback loop through the uncertainty. Before analyzing stability and performance, we must first ensure the well-posedness of the interconnection, such that solutions exist, are unique, and depend causally on inputs.

Definition 5. We say that the uncertain interconnection $[G, \Delta]$ is **well-posed** if:

- **Existence and Uniqueness:** For any $w \in \mathbf{L}_{e,[a,b]}$, there exist unique $z \in \mathbf{L}_{e,[a,b]}$, $z_\Delta \in \mathbf{L}_{e,[a,b]}$ and $w_\Delta \in \mathbf{L}_{e,[a,b]}$ such that, z, w, z_Δ, w_Δ satisfy the interconnection.
- **Causality:** If z_Δ, w_Δ and $\hat{z}_\Delta, \hat{w}_\Delta$ satisfy the interconnection, then $p_\tau(z_\Delta - \hat{z}_\Delta) = 0$ and $p_\tau(w_\Delta - \hat{w}_\Delta) = 0$, for all $\tau \geq 0$.

Having established the class of uncertain systems and their well-posedness, we now develop a framework to guarantee robust stability and performance for the class of systems defined in Definition 4.

B. Robust stability and performance

Traditional control system design often treats stability and performance shaping as separate objectives with distinct constraints. However, Integral Quadratic Constraints (IQCs) provide a unified framework where these trade-offs can be captured within a single constraint [3]. We now define what it means for the system to remain stable despite uncertainties. Robust stability requires that the system is well-posed for all admissible uncertainties and that the input-output map remains bounded. This involves removal of the performance channels in (3), and introducing auxiliary signals z_0 and w_0 .

Definition 6. The interconnection

$$z_\Delta = G_\Delta w_\Delta + z_0, \quad w_\Delta = \Delta z_\Delta + w_0,$$

with $z_0, w_0 \in \mathbf{L}_{e,[a,b]}$, is said to be **robustly stable** if:

- 1) The uncertain system is well-posed for all $\Delta \in \mathbf{\Delta}$.
- 2) The map from $\begin{bmatrix} z_0 \\ w_0 \end{bmatrix}$ to $\begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}$ is bounded on $\mathbf{L}_{[a,b]}$ for all $\Delta \in \mathbf{\Delta}$.

Within the IQC framework, the stability of an interconnection $[G, \Delta]$ is certified by verifying that the uncertainty operator Δ satisfies a quadratic constraint defined by an appropriate multiplier. This leads to the following formal definition

Definition 7. An uncertainty operator $\Delta : \mathbf{L}_{e,[a,b]}^{n_{z_\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{w_\Delta}}$ is said to satisfy the IQC defined by the operator $\Psi \in \mathbf{L}_{e,[a,b]}^{n_{w_\Delta} + n_{z_\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{w_\Delta} + n_{z_\Delta}}$ multiplier $\mathcal{M}_\Psi \in \mathcal{L}(\mathbf{L}_{[a,b]})$, parameterized by $\mathcal{V}_\Delta = \mathcal{V}_\Delta^* \in \mathcal{L}(\mathbb{R}L_2, \mathbb{R}L_2)$ if the inequality holds for all $z_\Delta \in \mathbf{L}_{e,[a,b]}$ and all $T \geq 0$,

$$\left\langle p_T \Psi \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \geq 0. \quad (4)$$

Moreover, let $\mathcal{V}_\Delta \{ \mathcal{Q}_\Delta, \mathcal{S}_\Delta, \mathcal{R}_\Delta \}$ be parameterized as,

$$\mathcal{V}_\Delta = \begin{bmatrix} \mathcal{Q}_\Delta & \mathcal{S}_\Delta \\ \mathcal{S}_\Delta^* & \mathcal{R}_\Delta \end{bmatrix}$$

Beyond stability, we are interested in ensuring that the system meets certain performance specifications in the presence of uncertainty. Robust performance extends the notion of robust stability by additionally requiring that a performance metric is satisfied.

Definition 8. **Robust performance** is achieved if the interconnection $[G, \Delta]$ is **robustly stable** for some performance multiplier $\mathcal{M}_{\mathcal{V}_p} \in \mathcal{L}(\mathbf{L}_{[a,b]})$, parameterized by $\mathcal{V}_p = \mathcal{V}_p^* \in \mathcal{L}(\mathbb{R}L_2, \mathbb{R}L_2)$ with $\mathcal{Q}_p \succeq 0$ such that, for all $w \in \mathbf{L}_{e,[a,b]}$, all $T \geq 0$, and some sufficiently small $\epsilon > 0$,

$$\left\langle p_T \begin{bmatrix} z \\ w \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_p} \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a,b]}}^2$$

Moreover, let $\mathcal{V}_p \{ \mathcal{Q}_p, \mathcal{S}_p, \mathcal{R}_p \}$ be parameterized as,

$$\mathcal{V}_p = \begin{bmatrix} \mathcal{Q}_p & \mathcal{S}_p \\ \mathcal{S}_p^* & \mathcal{R}_p \end{bmatrix}$$

In this framework, the multipliers $\mathcal{M}_\Psi$ and $\mathcal{M}_{\hat{\Psi}}$ serve distinct but complementary roles. The multiplier $\mathcal{M}_\Psi$ characterizes the behavior of uncertainties, nonlinearities, and other perturbations captured by the operator Δ , ensuring that the IQC is satisfied for the class of admissible disturbances. In contrast, the multiplier $\mathcal{M}_{\hat{\Psi}}$ encodes performance requirements for the nominal system, such as L_2 -gain bounds, passivity conditions, or other desired input-output properties. Together, these multipliers enable simultaneous verification of robust stability against perturbations and achievement of specified performance objectives.

C. Theorem for joint robust stability and performance

We now present the main theoretical result that provides conditions for simultaneous robust stability and performance verification.

Theorem 1 (Robust Stability and Performance). *Let $G, \Delta \in \Delta$ be given, and there exists $\Psi, \mathcal{M}_{\mathcal{V}_\Delta}$, such that,*

- *The interconnection $[G, \Delta]$ is well-posed for all $\Delta \in \Delta$*
- *The IQC in Definition 7 is satisfied for all $\Delta \in \Delta$.*

*Then the interconnection $[G, \Delta]$ is **robustly stable and robust performance** is achieved with respect to some $\mathcal{M}_{\mathcal{V}_p}$, in Definition 8, with $\mathcal{Q}_p \succeq 0$, if,*

$$\begin{aligned} & \left\langle p_T \hat{G} \begin{bmatrix} w_\Delta \\ w \end{bmatrix}, p_T \begin{bmatrix} \mathcal{M}_{\mathcal{V}_\Delta} & 0 \\ 0 & \mathcal{M}_{\mathcal{V}_p} \end{bmatrix} \hat{G} \begin{bmatrix} w_\Delta \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \\ & \leq -\epsilon (\|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2 + \|p_T w\|_{\mathbf{L}_{[a,b]}}^2) \end{aligned} \quad (5)$$

for all $w_\Delta \in \mathbf{L}_{e,[a,b]}^{n_{w_\Delta}}$, $w \in \mathbf{L}_{e,[a,b]}^{n_w}$ and $T \geq 0$, where $\hat{G} := \begin{bmatrix} \Psi & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} G_\Delta & G_{\Delta w} \\ -\frac{I}{G_{z\Delta}} & -\frac{0}{G_{zw}} \\ 0 & I \end{bmatrix}$ according to (3).

Proof. Suppose that (5) holds. This implies,

$$\begin{aligned} & \left\langle p_T \Psi \begin{bmatrix} G_\Delta \\ I \end{bmatrix} w_\Delta, p_T \mathcal{M}_{\mathcal{V}} \Psi \begin{bmatrix} G_\Delta \\ I \end{bmatrix} w_\Delta \right\rangle_{\mathbf{L}_{[a,b]}} \\ & + \left\langle p_T \begin{bmatrix} G_{z\Delta} \\ 0 \end{bmatrix} w_\Delta, p_T \mathcal{M}_{\hat{\mathcal{V}}} \begin{bmatrix} G_{z\Delta} \\ 0 \end{bmatrix} w_\Delta \right\rangle_{\mathbf{L}_{[a,b]}} \\ & \leq -\epsilon \|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2 \end{aligned}$$

Since $\hat{\mathcal{V}}_{11} \succeq 0$, we can then infer,

$$\left\langle p_T \Psi \begin{bmatrix} G_{\Delta w_\Delta} \\ w_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}} \Psi \begin{bmatrix} G_{\Delta w_\Delta} \\ w_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2$$

Then, using Theorem 6 in [3], we find that **robust stability** is guaranteed with respect to performance multiplier $\mathcal{M}_{\hat{\mathcal{V}}}$. Moreover, rewriting (5) as,

$$\begin{aligned} & \left\langle p_T \Psi \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}} \Psi \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} + \\ & \left\langle p_T \begin{bmatrix} z \\ w \end{bmatrix}, p_T \mathcal{M}_{\hat{\mathcal{V}}} \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon (\|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2 + \|p_T w\|_{\mathbf{L}_{[a,b]}}^2), \end{aligned}$$

and given that Ψ and $\mathcal{M}_{\mathcal{V}}$ satisfy the IQC in Definition 7 leads to,

$$\left\langle p_T \begin{bmatrix} z \\ w \end{bmatrix}, p_T \mathcal{M}_{\hat{\mathcal{V}}} \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w\|_{L_2}^2.$$

Thus by Definition 8 **robust performance** is achieved. ■

III. ESTABLISHED DUALITY

Theorem 2 (Dual Stability). *Suppose $\mathcal{T}, \mathcal{A} \in \mathcal{L}(\mathbb{R}L_2^{n_x})$ are PI operators. The following statements are equivalent.*

- $\lim_{t \rightarrow \infty} \mathcal{T}\mathbf{x}(t) \rightharpoonup 0$ for any $\mathbf{x}$ that satisfies $\partial_t(\mathcal{T}\mathbf{x}(t)) = \mathcal{A}\mathbf{x}(t)$ with initial condition $\mathbf{x}(0) \in \mathbb{R}L_2^{n_x}$.*
- $\lim_{t \rightarrow \infty} \mathcal{T}^*\underline{\mathbf{x}}(t) \rightharpoonup 0$ for any $\underline{\mathbf{x}}$ that satisfies $\mathcal{T}^*\underline{\mathbf{x}}(t) = \mathcal{A}^*\underline{\mathbf{x}}(t)$ with initial condition $\underline{\mathbf{x}}(0) \in \mathbb{R}L_2^{n_x}$.*

Note that " $\rightharpoonup 0$ " denotes weak convergence in the Hilbert space $\mathbb{R}L_2$.

Proof. To show sufficiency, suppose $\mathbf{x}$ satisfies $\partial_t(\mathcal{T}\mathbf{x}(t)) = \mathcal{A}\mathbf{x}(t)$ with initial condition $\mathbf{x}(0) \in \mathbb{R}L_2^{n_x}$ and $\lim_{t \rightarrow \infty} \mathcal{T}\mathbf{x}(t) \rightharpoonup 0$. Let $\underline{\mathbf{x}}$ satisfy $\mathcal{T}^*\underline{\mathbf{x}}(t) = \mathcal{A}^*\underline{\mathbf{x}}(t)$ with initial condition $\underline{\mathbf{x}}(0) \in \mathbb{R}L_2^{n_x}$. Then for any finite $t > 0$, using integration-by-parts, we get

$$\begin{aligned} & \int_0^t \langle \underline{\mathbf{x}}(t-s), \partial_s(\mathcal{T}\mathbf{x}(s)) \rangle_{\mathbb{R}L_2} ds \\ & = \langle \underline{\mathbf{x}}(0), \mathcal{T}\mathbf{x}(t) \rangle - \langle \underline{\mathbf{x}}(t), \mathcal{T}\mathbf{x}(0) \rangle \\ & - \int_0^t \langle \partial_s \underline{\mathbf{x}}(t-s), \mathcal{T}\mathbf{x}(s) \rangle ds. \end{aligned} \quad (6)$$

Then, we use a change of variable ($\theta = t - s$) on the last term in Eq. (6) to show

$$\begin{aligned} \int_0^t \langle \partial_s \underline{\mathbf{x}}(t-s), \mathcal{T}\mathbf{x}(s) \rangle ds &= \int_0^t \langle \dot{\underline{\mathbf{x}}}(\theta), \mathcal{T}\mathbf{x}(t-\theta) \rangle d\theta \\ &= \int_0^t \langle \mathcal{T}^* \dot{\underline{\mathbf{x}}}(\theta), \mathbf{x}(t-\theta) \rangle d\theta \end{aligned}$$

Furthermore, using the same variable change on the left-hand side of Eq. (6), we get

$$\begin{aligned} \int_0^t \langle \underline{\mathbf{x}}(t-s), \partial_s(\mathcal{T}\mathbf{x}(s)) \rangle_{\mathbb{R}L_2} ds \\ = \int_0^t \langle \underline{\mathbf{x}}(t-s), \mathcal{A}\mathbf{x}(s) \rangle ds = \int_0^t \langle \mathcal{A}^* \underline{\mathbf{x}}(\theta), \mathbf{x}(t-\theta) \rangle d\theta \end{aligned}$$

Substituting these two expressions into Eq. (6), we have

$$\begin{aligned} \int_0^t \langle \mathcal{A}^* \underline{\mathbf{x}}(\theta), \mathbf{x}(t-\theta) \rangle d\theta \\ = \langle \underline{\mathbf{x}}(0), \mathcal{T}\mathbf{x}(t) \rangle - \langle \underline{\mathbf{x}}(t), \mathcal{T}\mathbf{x}(0) \rangle \\ + \int_0^t \langle \mathcal{T}^* \dot{\underline{\mathbf{x}}}(\theta), \mathbf{x}(t-\theta) \rangle d\theta \end{aligned}$$

However, $\mathcal{A}^* \underline{\mathbf{x}}(\theta) = \mathcal{T}^* \dot{\underline{\mathbf{x}}}(\theta)$ for all $\theta \in [0, t]$, and hence

$$\langle \underline{\mathbf{x}}(0), \mathcal{T}\mathbf{x}(t) \rangle = \langle \mathcal{T}^* \underline{\mathbf{x}}(t), \mathbf{x}(0) \rangle, \text{ for all } t > 0. \quad (7)$$

Since $\lim_{t \rightarrow \infty} \mathcal{T}\mathbf{x}(t) = 0$, we have

$$\lim_{t \rightarrow \infty} \langle \mathcal{T}^* \underline{\mathbf{x}}(t), \mathbf{x}(0) \rangle = \lim_{t \rightarrow \infty} \langle \underline{\mathbf{x}}(0), \mathcal{T}\mathbf{x}(t) \rangle = 0.$$

Since $\mathbf{x}(0) \in \mathbb{R}L_2$ may be chosen arbitrarily and $\mathbb{R}L_2$ is Hilbert, this implies $\lim_{t \rightarrow \infty} \mathcal{T}^* \underline{\mathbf{x}}(t) \rightarrow 0$. Thus we have sufficiency. Since $\mathcal{T}^{**} = \mathcal{T}$ and existence of $\mathcal{T}^* \dot{\underline{\mathbf{x}}}(t)$ guarantees existence of $\partial_t(\mathcal{T}^* \underline{\mathbf{x}}(t))$, sufficiency implies necessity. $\blacksquare$

Theorem 3 (Dual L_2 -gain). Suppose $\mathcal{T}, \mathcal{A} \in \mathcal{L}(\mathbb{R}L_2^{\mathbf{n}_x})$, $\mathcal{B} \in \mathcal{L}(\mathbb{R}L_2^{\mathbf{n}_w}, \mathbb{R}L_2^{\mathbf{n}_x})$, $\mathcal{C} \in \mathcal{L}(\mathbb{R}L_2^{\mathbf{n}_x}, \mathbb{R}L_2^{\mathbf{n}_z})$, and $\mathcal{D} \in \mathcal{L}(\mathbb{R}L_2^{\mathbf{n}_w}, \mathbb{R}L_2^{\mathbf{n}_z})$ are PI operators. The following statements are equivalent.

a) For any $w(t) \in \mathbb{R}L_2^{\mathbf{n}_w}$, $x(t) \in \mathbb{R}L_2^{\mathbf{n}_x}$, and $z(t) \in \mathbb{R}L_2^{\mathbf{n}_z}$ that satisfy $\mathbf{x}(0) = 0$ and,

$$\begin{bmatrix} \partial_t(\mathcal{T}\mathbf{x})(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ w(t) \end{bmatrix} \quad (8)$$

we have that $\|z\|_{\mathbb{R}L_2} \leq \gamma \|w\|_{\mathbb{R}L_2}$.

b) For any $\underline{w}(t) \in \mathbb{R}L_2^{\mathbf{n}_z}$, $\underline{\mathbf{x}}(t) \in \mathbb{R}L_2^{\mathbf{n}_x}$, and $\underline{z}(t) \in \mathbb{R}L_2^{\mathbf{n}_w}$ that satisfy $\underline{\mathbf{x}}(0) = 0$ and,

$$\begin{bmatrix} \mathcal{T}^* \dot{\underline{\mathbf{x}}}(t) \\ \underline{z}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* & \mathcal{C}^* \\ \mathcal{B}^* & \mathcal{D}^* \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}(t) \\ \underline{w}(t) \end{bmatrix} \quad (9)$$

we have that $\|\underline{z}\|_{\mathbb{R}L_2} \leq \gamma \|\underline{w}\|_{\mathbb{R}L_2}$.

Proof. To show sufficiency, let $\mathbf{x}(t) \in \mathbb{R}L_2^{\mathbf{n}_x}$ and $z(t) \in \mathbb{R}L_2^{\mathbf{n}_z}$ satisfy Eq. 8 for $\mathbf{x}(0) = 0$ and some $w(t) \in \mathbb{R}L_2^{\mathbf{n}_w}$. Then $\|z(t)\|_{\mathbb{R}L_2} \leq \gamma \|w(t)\|_{\mathbb{R}L_2}$. Let $\underline{\mathbf{x}}(t) \in \mathbb{R}L_2^{\mathbf{n}_x}$ and $\underline{w}(t) \in \mathbb{R}L_2^{\mathbf{n}_z}$ satisfy Eq. (9) for $\underline{\mathbf{x}}(0) = 0$ and some $\underline{w}(t) \in \mathbb{R}L_2^{\mathbf{n}_z}$. Then, as in Eq. 6 in the proof of Theorem 2 and substituting initial conditions, we find

$$\int_0^t \langle \underline{\mathbf{x}}(t-s), \partial_s(\mathcal{T}\mathbf{x}(s)) \rangle ds = \int_0^t \langle \mathcal{T}^* \dot{\underline{\mathbf{x}}}(\theta), \mathbf{x}(t-\theta) \rangle d\theta \quad (10)$$

Furthermore, by using the variable change $\theta = t - s$ on the left-hand side of the above equation,

$$\begin{aligned} \int_0^t \langle \underline{\mathbf{x}}(t-s), \partial_s(\mathcal{T}\mathbf{x}(s)) \rangle ds \\ = \int_0^t \langle \underline{\mathbf{x}}(t-s), \mathcal{A}\mathbf{x}(s) \rangle ds + \int_0^t \langle \underline{\mathbf{x}}(t-s), \mathcal{B}w(s) \rangle ds \\ = \int_0^t \langle \mathcal{A}^* \underline{\mathbf{x}}(\theta), \mathbf{x}(t-\theta) \rangle d\theta + \int_0^t \langle \mathcal{B}^* \underline{\mathbf{x}}(\theta), w(t-\theta) \rangle d\theta \end{aligned} \quad (11)$$

Combining Eq. (10) and (11), we obtain

$$\begin{aligned} & \int_0^t \langle \mathcal{T}^* \dot{\underline{\mathbf{x}}}(\theta), \mathbf{x}(t - \theta) \rangle d\theta \\ &= \int_0^t \langle \mathcal{A}^* \underline{\mathbf{x}}(\theta), \mathbf{x}(t - \theta) \rangle d\theta + \int_0^t \langle \mathcal{B}^* \underline{\mathbf{x}}(\theta), w(t - \theta) \rangle d\theta \end{aligned}$$

However, $\mathcal{T}^* \dot{\underline{\mathbf{x}}}(t) - \mathcal{A}^* \underline{\mathbf{x}}(t) = \mathcal{C}^* \underline{w}(t)$, so

$$\begin{aligned} & \int_0^t \langle \mathcal{C}^* \underline{w}(\theta), \mathbf{x}(t - \theta) \rangle d\theta \\ &= \int_0^t \langle \mathcal{T}^* \dot{\underline{\mathbf{x}}}(\theta) - \mathcal{A}^* \underline{\mathbf{x}}(\theta), \mathbf{x}(t - \theta) \rangle d\theta \\ &= \int_0^t \langle \mathcal{B}^* \underline{\mathbf{x}}(\theta), w(t - \theta) \rangle d\theta. \end{aligned}$$

Since $z(t) = \mathcal{C}\mathbf{x}(t) + \mathcal{D}w(t)$, we obtain

$$\begin{aligned} & \int_0^t \langle \underline{w}(\theta), z(t - \theta) \rangle d\theta - \int_0^t \langle \underline{w}(\theta), \mathcal{D}w(t - \theta) \rangle d\theta \\ &= \int_0^t \langle \underline{w}(\theta), \mathcal{C}\mathbf{x}(t - \theta) \rangle d\theta = \int_0^t \langle \mathcal{C}^* \underline{w}(\theta), \mathbf{x}(t - \theta) \rangle d\theta \\ &= \int_0^t \langle \mathcal{B}^* \underline{\mathbf{x}}(\theta), w(t - \theta) \rangle d\theta. \end{aligned}$$

Likewise, we know $\underline{z}(t) = \mathcal{B}^* \underline{\mathbf{x}}(t) + \mathcal{D}^* \underline{w}(t)$. Hence

$$\begin{aligned} & \int_0^t \langle \underline{z}(\theta), w(t - \theta) \rangle d\theta - \int_0^t \langle \mathcal{D}^* \underline{w}(\theta), w(t - \theta) \rangle d\theta \\ &= \int_0^t \langle \mathcal{B}^* \underline{\mathbf{x}}(\theta), w(t - \theta) \rangle d\theta \\ &= \int_0^t \langle \underline{w}(\theta), z(t - \theta) \rangle d\theta - \int_0^t \langle \underline{w}(\theta), \mathcal{D}w(t - \theta) \rangle d\theta \end{aligned}$$

We conclude that for any $t > 0$, if $z(t)$ and $w(t)$ satisfy the primal PIE and $\underline{z}(t)$ and $\underline{w}(t)$ satisfy the dual PIE, then

$$\int_0^t \langle \underline{z}(\theta), w(t - \theta) \rangle d\theta = \int_0^t \langle \underline{w}(\theta), z(t - \theta) \rangle d\theta. \quad (12)$$

Now, for any $\underline{w}(t) \in \mathbb{R}L_2$, let $\underline{z}(t)$ solve the dual PIE for some $\underline{\mathbf{x}}(t)$. For any fixed $\tau > 0$, define $w(t) = \underline{z}(\tau - t)$ for $t \leq \tau$ and $w(t) = 0$ for $t > \tau$. Then $w(t) \in \mathbb{R}L_2$ and for this input, let $z(t)$ solve the primal PIE for some $\underline{\mathbf{x}}(t)$. Then, if we define the truncation operator P_τ , we have

$$\begin{aligned} \|P_\tau \underline{z}(t)\|_{\mathbb{R}L_2}^2 &= \int_0^\tau \langle \underline{z}(s), \underline{z}(s) \rangle ds = \int_0^\tau \langle \underline{z}(s), w(\tau - s) \rangle ds \\ &= \int_0^\tau \langle \underline{w}(s), z(\tau - s) \rangle ds \\ &\leq \|P_\tau \underline{w}(t)\|_{\mathbb{R}L_2} \|P_\tau z(t)\|_{\mathbb{R}L_2} \\ &\leq \|P_\tau \underline{w}(t)\|_{\mathbb{R}L_2} \|z(t)\|_{\mathbb{R}L_2} \\ &\leq \gamma \|P_\tau \underline{w}(t)\|_{\mathbb{R}L_2} \|w(t)\|_{\mathbb{R}L_2} \\ &= \gamma \|P_\tau \underline{w}(t)\|_{\mathbb{R}L_2} \|P_\tau w(t)\|_{\mathbb{R}L_2} \\ &= \gamma \|P_\tau \underline{w}(t)\|_{\mathbb{R}L_2} \|P_\tau \underline{z}(t)\|_{\mathbb{R}L_2}. \end{aligned}$$

Therefore, we have that $\|P_\tau \underline{z}(t)\|_{\mathbb{R}L_2} \leq \gamma \|P_\tau \underline{w}(t)\|_{\mathbb{R}L_2}$ for all $\tau \geq 0$. We conclude that $\|\underline{z}(t)\|_{\mathbb{R}L_2} \leq \gamma \|\underline{w}(t)\|_{\mathbb{R}L_2}$. Since $\mathcal{T}^{**} = \mathcal{T}$ and

$$\begin{bmatrix} (\mathcal{A}^*)^* & (\mathcal{B}^*)^* \\ (\mathcal{C}^*)^* & (\mathcal{D}^*)^* \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* & \mathcal{C}^* \\ \mathcal{B}^* & \mathcal{D}^* \end{bmatrix}^* = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix}^{**} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix}$$

we have that sufficiency implies necessity. ■

IV. DUALITY DERIVATIONS FOR DYNAMIC IQCs

Definition 9 (Dual Realization). For a PIE system with corresponding PI operators $G\{\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}\}$ the state space representation is defined as

$$\begin{bmatrix} \mathcal{T}\dot{\mathbf{x}}(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ u(t) \end{bmatrix}, \quad (13)$$

define the dual realization, Theorem 3, by

$$G^\sharp := G\{\mathcal{T}^*, \mathcal{A}^*, \mathcal{C}^*, \mathcal{B}^*, \mathcal{D}^*\}. \quad (14)$$

Definition 10. For $w(t) \in \mathbb{R}L_2^{\mathbf{n}_w}$, $\mathbf{x}(t) \in \mathbb{R}L_2^{\mathbf{n}_x}$, and $z(t) \in \mathbb{R}L_2^{\mathbf{n}_z}$ that satisfy $\mathbf{x}(0) = 0$, the **Primal System** $G\{\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}\}$ is defined as,

$$\begin{bmatrix} \mathcal{T}\dot{\mathbf{x}}(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ w(t) \end{bmatrix} \quad (15)$$

where $\mathcal{T}, \mathcal{A} \in \Pi_4^{\mathbf{n}_x \times \mathbf{n}_x}$, $\mathcal{B} \in \Pi_4^{\mathbf{n}_x \times \mathbf{n}_w}$, $\mathcal{C} \in \Pi_4^{\mathbf{n}_z \times \mathbf{n}_x}$, and $\mathcal{D} \in \Pi_4^{\mathbf{n}_z \times \mathbf{n}_w}$ are PI operators. The **Scaled Primal System** is expressed in terms of $\tilde{w}(t) \in \mathbb{R}L_2^{\mathbf{n}_w}$ and $\tilde{z}(t) \in \mathbb{R}L_2^{\mathbf{n}_z}$, which satisfy,

$$\Psi \begin{bmatrix} z(t) \\ w(t) \end{bmatrix} = \begin{bmatrix} \Psi_z & \Psi_{zw} \\ \Psi_{wz} & \Psi_w \end{bmatrix} \begin{bmatrix} z(t) \\ w(t) \end{bmatrix} = \begin{bmatrix} \tilde{z}(t) \\ \tilde{w}(t) \end{bmatrix}. \quad (16)$$

The operator $\Psi \in \mathcal{L}(\mathbb{R}L_2)$ is called the **Primal Scaling** of the system. The state space representation of (16) is given by

$$\begin{bmatrix} \mathcal{T}_\Psi \mathbf{x}_\Psi(t) \\ \tilde{z}(t) \\ \tilde{w}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_\Psi & \mathcal{B}_\Psi^z & \mathcal{B}_\Psi^w \\ \mathcal{C}_\Psi^z & \mathcal{D}_\Psi^z & \mathcal{D}_\Psi^{zw} \\ \mathcal{C}_\Psi^w & \mathcal{D}_\Psi^{wz} & \mathcal{D}_\Psi^w \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Psi(t) \\ z(t) \\ w(t) \end{bmatrix} \quad (17)$$

where the entries are PI operators of appropriate dimension.

Definition 11. Let $\Psi = \begin{bmatrix} \Psi_z & \Psi_{zw} \\ \Psi_{wz} & \Psi_w \end{bmatrix}$ be an invertible scaling with state-space representation (17), and let $\Psi^\sharp$ be defined by the dual realization. Let $\hat{\mathcal{J}} := \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$. Define the dual scaling operation

$$\mathbf{D}(\Psi) := (\hat{\mathcal{J}}^* \Psi^\sharp \hat{\mathcal{J}})^{-1} = \begin{bmatrix} \Psi_w^\sharp & -\Psi_{zw}^\sharp \\ -\Psi_{wz}^\sharp & \Psi_z^\sharp \end{bmatrix}^{-1}. \quad (18)$$

Since $\hat{\mathcal{J}}^* = \hat{\mathcal{J}}^{-1}$, $\hat{\mathcal{J}}^* = -\hat{\mathcal{J}}$, $(\Psi^\sharp)^\sharp = \Psi$, and $(\Psi^{-1})^{-1} = \Psi$. Then, for any factorization $\Psi = UV$ in which U and V are invertible, the dual scaling operation is multiplicative,

$$\begin{aligned} \mathbf{D}(UV) &= (\hat{\mathcal{J}}^* V^\sharp U^\sharp \hat{\mathcal{J}})^{-1} \\ &= (\hat{\mathcal{J}}^* V^\sharp \hat{\mathcal{J}} \hat{\mathcal{J}}^* U^\sharp \hat{\mathcal{J}})^{-1} \\ &= \mathbf{D}(U) \mathbf{D}(V), \end{aligned} \quad (19)$$

and the dual scaling operation is an involution,

$$\mathbf{D}(\mathbf{D}(\Psi)) = \Psi. \quad (20)$$

Definition 12. For $\underline{w}(t) \in \mathbb{R}L_2^{\mathbf{n}_z}$, $\underline{\mathbf{x}}(t) \in \mathbb{R}L_2^{\mathbf{n}_x}$, and $\underline{z}(t) \in \mathbb{R}L_2^{\mathbf{n}_w}$ that satisfy $\underline{\mathbf{x}}(0) = 0$, the **Dual System** $G^\sharp$ is defined by applying the dual realization to the **Primal System**:

$$\begin{bmatrix} \mathcal{T}^* \dot{\underline{\mathbf{x}}}(t) \\ \underline{z}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* & \mathcal{C}^* \\ \mathcal{B}^* & \mathcal{D}^* \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}(t) \\ \underline{w}(t) \end{bmatrix}. \quad (21)$$

The **Scaled Dual System** is expressed in terms of $\underline{\tilde{w}}(t) \in \mathbb{R}L_2^{\mathbf{n}_z}$ and $\underline{\tilde{z}}(t) \in \mathbb{R}L_2^{\mathbf{n}_w}$, which satisfy,

$$\mathbf{D}(\Psi) \begin{bmatrix} \underline{z}(t) \\ \underline{w}(t) \end{bmatrix} = \begin{bmatrix} \underline{\tilde{z}}(t) \\ \underline{\tilde{w}}(t) \end{bmatrix}. \quad (22)$$

The operator $\mathbf{D}(\Psi) \in \mathcal{L}(\mathbb{R}L_2)$ is called the **Dual Scaling** of the system and is the dual of the **Primal Scaling**. Thus the state space representation of (22) is given by

$$\begin{bmatrix} \mathcal{T}_\Psi^* \dot{\underline{\mathbf{x}}}_\Psi(t) \\ \underline{z}(t) \\ \underline{w}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_\Psi^* & -\mathcal{C}_\Psi^{w*} & \mathcal{C}_\Psi^{z*} \\ -\mathcal{B}_\Psi^{w*} & \mathcal{D}_\Psi^{ww*} & -\mathcal{D}_\Psi^{zw*} \\ \mathcal{B}_\Psi^{z*} & -\mathcal{D}_\Psi^{wz*} & \mathcal{D}_\Psi^{zz*} \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}_\Psi(t) \\ \underline{\tilde{z}}(t) \\ \underline{\tilde{w}}(t) \end{bmatrix} \quad (23)$$

Adopted from [4]

Definition 13. (*Potapov–Ginzburg Transform*) Let $\Psi : \begin{bmatrix} z \\ w \end{bmatrix} \rightarrow \begin{bmatrix} \tilde{z} \\ \tilde{w} \end{bmatrix}$, where $\Psi = \begin{bmatrix} \Psi_z & \Psi_{zw} \\ \Psi_{wz} & \Psi_w \end{bmatrix}$ and Ψ_w is invertible. Then the Potapov–Ginzburg transform of Ψ is defined as

$$\begin{bmatrix} \tilde{z} \\ w \end{bmatrix} = \Psi^\dagger \begin{bmatrix} \tilde{w} \\ z \end{bmatrix}, \text{ where } \Psi^\dagger = \begin{bmatrix} \Psi_{zw}\Psi_w^{-1} & \Psi_z - \Psi_{zw}\Psi_w^{-1}\Psi_{wz} \\ \Psi_w^{-1} & -\Psi_w^{-1}\Psi_{wz} \end{bmatrix}. \quad (24)$$

Assumed to admit the state-space representation

$$\begin{bmatrix} \mathcal{T}_\Psi \dot{\mathbf{x}}_\Psi(t) \\ \tilde{z}(t) \\ w(t) \end{bmatrix} = \begin{bmatrix} \hat{\mathcal{A}}_\Psi & \hat{\mathcal{B}}_\Psi^{\tilde{w}} & \hat{\mathcal{B}}_\Psi^z \\ \hat{\mathcal{C}}_\Psi^z & \hat{\mathcal{D}}_\Psi^{z\tilde{w}} & \hat{\mathcal{D}}_\Psi^z \\ \hat{\mathcal{C}}_\Psi^w & \hat{\mathcal{D}}_\Psi^{w\tilde{w}} & \hat{\mathcal{D}}_\Psi^{wz} \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Psi(t) \\ \tilde{w}(t) \\ z(t) \end{bmatrix}, \quad (25)$$

where

$$\begin{aligned} \hat{\mathcal{A}}_\Psi &:= \mathcal{A}_\Psi - \mathcal{B}_\Psi^w (\mathcal{D}_\Psi^w)^{-1} \mathcal{C}_\Psi^w, & \hat{\mathcal{B}}_\Psi^{\tilde{w}} &:= \mathcal{B}_\Psi^w (\mathcal{D}_\Psi^w)^{-1}, & \hat{\mathcal{B}}_\Psi^z &:= \mathcal{B}_\Psi^z - \mathcal{B}_\Psi^w (\mathcal{D}_\Psi^w)^{-1} \mathcal{D}_\Psi^{wz}, & \hat{\mathcal{C}}_\Psi^z &:= \mathcal{C}_\Psi^z - \mathcal{D}_\Psi^{zw} (\mathcal{D}_\Psi^w)^{-1} \mathcal{C}_\Psi^w, \\ \hat{\mathcal{D}}_\Psi^z &:= \mathcal{D}_\Psi^z - \mathcal{D}_\Psi^{zw} (\mathcal{D}_\Psi^w)^{-1} \mathcal{D}_\Psi^{wz}, & \hat{\mathcal{D}}_\Psi^{z\tilde{w}} &:= \mathcal{D}_\Psi^{zw} (\mathcal{D}_\Psi^w)^{-1}, & \hat{\mathcal{C}}_\Psi^w &:= -(\mathcal{D}_\Psi^w)^{-1} \mathcal{C}_\Psi^w, & \hat{\mathcal{D}}_\Psi^{w\tilde{w}} &:= (\mathcal{D}_\Psi^w)^{-1}, \\ \hat{\mathcal{D}}_\Psi^{wz} &:= -(\mathcal{D}_\Psi^w)^{-1} \mathcal{D}_\Psi^{wz}. \end{aligned}$$

Lemma 1. (*Potapov–Ginzburg Transform Equivalence*) The **Scaled Primal System** (respectively, **Scaled Dual System**) is equivalent, under an invertible congruence transformation, to the **Primal System** (respectively, **Dual System**) augmented with the Potapov–Ginzburg transform of the **Primal Scaling** (respectively, **Dual Scaling**).

Proof. We first show the primal derivation explicitly. To show sufficiency, first augment the **Primal System** with the Potapov–Ginzburg transform of the **Primal Scaling**. Substitution of $z = \mathcal{C}\mathbf{x} + \mathcal{D}w$ into (25) gives the algebraic loop

$$w = \Lambda_p \hat{\mathcal{D}}_\Psi^{wz} \mathcal{C}\mathbf{x} + \Lambda_p \hat{\mathcal{C}}_\Psi^w \mathbf{x}_\Psi + \Lambda_p \hat{\mathcal{D}}_\Psi^{w\tilde{w}} \tilde{w}.$$

Assume that this loop is well-posed, and define

$$\Lambda_p := (I - \hat{\mathcal{D}}_\Psi^{wz} \mathcal{D})^{-1}, \quad \Lambda_z := (I - \mathcal{D} \hat{\mathcal{D}}_\Psi^{wz})^{-1}.$$

Solving for w and substituting into the plant state equation gives

$$\mathcal{T} \dot{\mathbf{x}} = \left(\mathcal{A} + \mathcal{B} \Lambda_p \hat{\mathcal{D}}_\Psi^{wz} \mathcal{C} \right) \mathbf{x} + \mathcal{B} \Lambda_p \hat{\mathcal{C}}_\Psi^w \mathbf{x}_\Psi + \mathcal{B} \Lambda_p \hat{\mathcal{D}}_\Psi^{w\tilde{w}} \tilde{w}.$$

Using $\mathcal{D} \Lambda_p = \Lambda_z \mathcal{D}$ and $I + \mathcal{D} \Lambda_p \hat{\mathcal{D}}_\Psi^{wz} = \Lambda_z$, as verified below in the push-through identities, (27) and (30), we obtain

$$z = \Lambda_z \mathcal{C} \mathbf{x} + \Lambda_z \mathcal{D} \hat{\mathcal{C}}_\Psi^w \mathbf{x}_\Psi + \Lambda_z \mathcal{D} \hat{\mathcal{D}}_\Psi^{w\tilde{w}} \tilde{w}.$$

Substitution into the equations for $\dot{\mathbf{x}}_\Psi$ and $\tilde{z}$ yields the augmented realization

$$\begin{bmatrix} \mathcal{T} \dot{\mathbf{x}}(t) \\ \mathcal{T}_\Psi \dot{\mathbf{x}}_\Psi(t) \\ \tilde{z}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} + \mathcal{B} \Lambda_p \hat{\mathcal{D}}_\Psi^{wz} \mathcal{C} & \mathcal{B} \Lambda_p \hat{\mathcal{C}}_\Psi^w & \mathcal{B} \Lambda_p \hat{\mathcal{D}}_\Psi^{w\tilde{w}} \\ \hat{\mathcal{B}}_\Psi^z \Lambda_z \mathcal{C} & \hat{\mathcal{A}}_\Psi + \hat{\mathcal{B}}_\Psi^z \Lambda_z \mathcal{D} \hat{\mathcal{C}}_\Psi^w & \hat{\mathcal{B}}_\Psi^{\tilde{w}} + \hat{\mathcal{B}}_\Psi^z \Lambda_z \mathcal{D} \hat{\mathcal{D}}_\Psi^{w\tilde{w}} \\ \hat{\mathcal{D}}_\Psi^z \Lambda_z \mathcal{C} & \hat{\mathcal{C}}_\Psi^z + \hat{\mathcal{D}}_\Psi^z \Lambda_z \mathcal{D} \hat{\mathcal{C}}_\Psi^w & \hat{\mathcal{D}}_\Psi^{z\tilde{w}} + \hat{\mathcal{D}}_\Psi^z \Lambda_z \mathcal{D} \hat{\mathcal{D}}_\Psi^{w\tilde{w}} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{x}_\Psi(t) \\ \tilde{w}(t) \end{bmatrix} \quad (26)$$

Define also

$$\Lambda_c := (\mathcal{D}_\Psi^{wz} \mathcal{D} + \mathcal{D}_\Psi^w)^{-1} = (I + (\mathcal{D}_\Psi^w)^{-1} \mathcal{D}_\Psi^{wz} \mathcal{D})^{-1} (\mathcal{D}_\Psi^w)^{-1} = (\mathcal{D}_\Psi^w)^{-1} (I + \mathcal{D}_\Psi^{wz} \mathcal{D} (\mathcal{D}_\Psi^w)^{-1})^{-1}.$$

The push-through identities

$$(I + \mathcal{U}\mathcal{V})^{-1} \mathcal{U} = \mathcal{U} (I + \mathcal{V}\mathcal{U})^{-1}, \quad (I + \mathcal{U}\mathcal{V})^{-1} = I - \mathcal{U} (I + \mathcal{V}\mathcal{U})^{-1} \mathcal{V},$$

give the following relations

$$\Lambda_p(\mathcal{D}_\Psi^w)^{-1} = (I + (\mathcal{D}_\Psi^w)^{-1}\mathcal{D}_\Psi^{wz}\mathcal{D})^{-1}(\mathcal{D}_\Psi^w)^{-1} = \Lambda_c \quad (27)$$

$$\begin{aligned} (\mathcal{D}_\Psi^w)^{-1}\mathcal{D}_\Psi^{wz}\Lambda_z &= (\mathcal{D}_\Psi^w)^{-1}\mathcal{D}_\Psi^{wz}(I + \mathcal{D}(\mathcal{D}_\Psi^w)^{-1}\mathcal{D}_\Psi^{wz})^{-1} \\ &= (\mathcal{D}_\Psi^w)^{-1}(I + \mathcal{D}_\Psi^{wz}\mathcal{D}(\mathcal{D}_\Psi^w)^{-1})^{-1}\mathcal{D}_\Psi^{wz} = \Lambda_c\mathcal{D}_\Psi^{wz} \end{aligned} \quad (28)$$

$$\begin{aligned} \Lambda_z\mathcal{D}(\mathcal{D}_\Psi^w)^{-1} &= (I + \mathcal{D}(\mathcal{D}_\Psi^w)^{-1}\mathcal{D}_\Psi^{wz})^{-1}\mathcal{D}(\mathcal{D}_\Psi^w)^{-1} \\ &= \mathcal{D}(\mathcal{D}_\Psi^w)^{-1}(I + \mathcal{D}_\Psi^{wz}\mathcal{D}(\mathcal{D}_\Psi^w)^{-1})^{-1} = \mathcal{D}\Lambda_c \end{aligned} \quad (29)$$

$$\begin{aligned} \Lambda_z &= (I + \mathcal{D}(\mathcal{D}_\Psi^w)^{-1}\mathcal{D}_\Psi^{wz})^{-1} \\ &= I - \mathcal{D}(\mathcal{D}_\Psi^w)^{-1}(I + \mathcal{D}_\Psi^{wz}\mathcal{D}(\mathcal{D}_\Psi^w)^{-1})^{-1}\mathcal{D}_\Psi^{wz} \\ &= I - \mathcal{D}\Lambda_c\mathcal{D}_\Psi^{wz} \end{aligned} \quad (30)$$

$$\begin{aligned} (\mathcal{D}_\Psi^w)^{-1} - \Lambda_c\mathcal{D}_\Psi^{wz}\mathcal{D}(\mathcal{D}_\Psi^w)^{-1} &= (I - \Lambda_c\mathcal{D}_\Psi^{wz}\mathcal{D})(\mathcal{D}_\Psi^w)^{-1} \\ &= \Lambda_c(\mathcal{D}_\Psi^w + \mathcal{D}_\Psi^{wz}\mathcal{D} - \mathcal{D}_\Psi^{wz}\mathcal{D})(\mathcal{D}_\Psi^w)^{-1} \\ &= \Lambda_c. \end{aligned} \quad (31)$$

We now express the blocks of (26) in terms of Λ_c . For the first row, direct substitution and (27) give

$$[\mathcal{A} + \mathcal{B}\Lambda_p\hat{\mathcal{D}}_\Psi^{wz}\mathcal{C} \quad \mathcal{B}\Lambda_p\hat{\mathcal{C}}_\Psi^w \quad \mathcal{B}\Lambda_p\hat{\mathcal{D}}_\Psi^{w\tilde{w}}] = [\mathcal{A} - \mathcal{B}\Lambda_c\mathcal{D}_\Psi^{wz}\mathcal{C} \quad -\mathcal{B}\Lambda_c\mathcal{C}_\Psi^w \quad \mathcal{B}\Lambda_c].$$

For the second row, (28) and (30) give

$$\begin{aligned} \hat{\mathcal{B}}_\Psi^z\Lambda_z &= (\mathcal{B}_\Psi^z - \mathcal{B}_\Psi^w(\mathcal{D}_\Psi^w)^{-1}\mathcal{D}_\Psi^{wz})\Lambda_z \\ &= \mathcal{B}_\Psi^z\Lambda_z - \mathcal{B}_\Psi^w(\mathcal{D}_\Psi^w)^{-1}\mathcal{D}_\Psi^{wz}\Lambda_z \\ &= \mathcal{B}_\Psi^z(I - \mathcal{D}\Lambda_c\mathcal{D}_\Psi^{wz}) - \mathcal{B}_\Psi^w\Lambda_c\mathcal{D}_\Psi^{wz} \\ &= \mathcal{B}_\Psi^z - (\mathcal{B}_\Psi^z\mathcal{D} + \mathcal{B}_\Psi^w)\Lambda_c\mathcal{D}_\Psi^{wz}. \end{aligned}$$

Its third entry follows from (31):

$$\begin{aligned} \hat{\mathcal{B}}_\Psi^{\tilde{w}} + \hat{\mathcal{B}}_\Psi^z\Lambda_z\mathcal{D}\hat{\mathcal{D}}_\Psi^{w\tilde{w}} &= \mathcal{B}_\Psi^w(\mathcal{D}_\Psi^w)^{-1} + [\mathcal{B}_\Psi^z - (\mathcal{B}_\Psi^z\mathcal{D} + \mathcal{B}_\Psi^w)\Lambda_c\mathcal{D}_\Psi^{wz}]\mathcal{D}(\mathcal{D}_\Psi^w)^{-1} \\ &= (\mathcal{B}_\Psi^z\mathcal{D} + \mathcal{B}_\Psi^w)[(\mathcal{D}_\Psi^w)^{-1} - \Lambda_c\mathcal{D}_\Psi^{wz}\mathcal{D}(\mathcal{D}_\Psi^w)^{-1}] \\ &= (\mathcal{B}_\Psi^z\mathcal{D} + \mathcal{B}_\Psi^w)\Lambda_c. \end{aligned}$$

The third row is obtained analogously:

$$\hat{\mathcal{D}}_\Psi^z\Lambda_z = \mathcal{D}_\Psi^z - (\mathcal{D}_\Psi^z\mathcal{D} + \mathcal{D}_\Psi^{zw})\Lambda_c\mathcal{D}_\Psi^{wz} \quad \hat{\mathcal{D}}_\Psi^{z\tilde{w}} + \hat{\mathcal{D}}_\Psi^z\Lambda_z\mathcal{D}\hat{\mathcal{D}}_\Psi^{w\tilde{w}} = (\mathcal{D}_\Psi^z\mathcal{D} + \mathcal{D}_\Psi^{zw})\Lambda_c.$$

Collecting these blocks and retaining $\tilde{w}$ as an output yields

$$\begin{bmatrix} \mathcal{T}\dot{\mathbf{x}}(t) \\ \mathcal{T}_\Psi\dot{\mathbf{x}}_\Psi(t) \\ \tilde{z}(t) \\ \tilde{w}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} - \mathcal{B}\Lambda_c\mathcal{D}_\Psi^{wz}\mathcal{C} & -\mathcal{B}\Lambda_c\mathcal{C}_\Psi^w & \mathcal{B}\Lambda_c \\ \mathcal{B}_\Psi^z\mathcal{C} - (\mathcal{B}_\Psi^z\mathcal{D} + \mathcal{B}_\Psi^w)\Lambda_c\mathcal{D}_\Psi^{wz}\mathcal{C} & \mathcal{A}_\Psi - (\mathcal{B}_\Psi^z\mathcal{D} + \mathcal{B}_\Psi^w)\Lambda_c\mathcal{C}_\Psi^w & (\mathcal{B}_\Psi^z\mathcal{D} + \mathcal{B}_\Psi^w)\Lambda_c \\ \mathcal{D}_\Psi^z\mathcal{C} - (\mathcal{D}_\Psi^z\mathcal{D} + \mathcal{D}_\Psi^{zw})\Lambda_c\mathcal{D}_\Psi^{wz}\mathcal{C} & \mathcal{C}_\Psi^z - (\mathcal{D}_\Psi^z\mathcal{D} + \mathcal{D}_\Psi^{zw})\Lambda_c\mathcal{C}_\Psi^w & (\mathcal{D}_\Psi^z\mathcal{D} + \mathcal{D}_\Psi^{zw})\Lambda_c \\ 0 & 0 & I \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{x}_\Psi(t) \\ \tilde{w}(t) \end{bmatrix} \quad (32)$$

The corresponding invertible congruence transformation is

$$\begin{bmatrix} \mathbf{x}(t) \\ \mathbf{x}_\Psi(t) \\ \tilde{w}(t) \end{bmatrix} = \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ \mathcal{D}_\Psi^{wz}\mathcal{C} & \mathcal{C}_\Psi^w & \Lambda_c^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{x}_\Psi(t) \\ w(t) \end{bmatrix} \quad \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{x}_\Psi(t) \\ w(t) \end{bmatrix} = \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ -\Lambda_c\mathcal{D}_\Psi^{wz}\mathcal{C} & -\Lambda_c\mathcal{C}_\Psi^w & \Lambda_c \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{x}_\Psi(t) \\ \tilde{w}(t) \end{bmatrix} \quad (33)$$

Applying (33) to (32) gives

$$\begin{bmatrix} \mathcal{T}\dot{\mathbf{x}}(t) \\ \mathcal{T}_\Psi\dot{\mathbf{x}}_\Psi(t) \\ \tilde{z}(t) \\ \tilde{w}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & 0 & \mathcal{B} \\ \mathcal{B}_\Psi^z\mathcal{C} & \mathcal{A}_\Psi & \mathcal{B}_\Psi^z\mathcal{D} + \mathcal{B}_\Psi^w \\ \mathcal{D}_\Psi^z\mathcal{C} & \mathcal{C}_\Psi^z & \mathcal{D}_\Psi^z\mathcal{D} + \mathcal{D}_\Psi^{zw} \\ \mathcal{D}_\Psi^{wz}\mathcal{C} & \mathcal{C}_\Psi^w & \mathcal{D}_\Psi^{wz}\mathcal{D} + \mathcal{D}_\Psi^w \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{x}_\Psi(t) \\ w(t) \end{bmatrix}. \quad (34)$$

The realization in (34) is the direct augmentation of the **Primal System** with the **Primal Scaling**, and hence is the **Scaled Primal System**. For necessity, applying (33) to (34) recovers (32), and reversing (27)–(31) recovers (26). The dual case follows by applying the same argument to the **Dual System** and the **Dual Scaling**. $\blacksquare$

Theorem 4 (Scaled Dual L_2 -gain). *Assume well-posedness. The following statements are equivalent.*

- 1) *For all $\tilde{w} \in \mathbb{R}L_2^{\mathbf{n}_w}$ and $\mathbf{x}(0) = 0$, the **Scaled Primal System** satisfies $\|P_\tau \tilde{z}\| \leq \|P_\tau \tilde{w}\|$ for all $\tau \geq 0$.*
- 2) *For all $\tilde{w} \in \mathbb{R}L_2^{\mathbf{n}_z}$ and $\mathbf{x}(0) = 0$, the **Scaled Dual System** satisfies $\|P_\tau \tilde{z}\| \leq \|P_\tau \tilde{w}\|$ for all $\tau \geq 0$.*

Proof. The Potapov-Ginzburg transformation of (23) gives

$$\begin{aligned}\tilde{z} &= (\mathcal{D}_\Psi^*)^{-1} \mathcal{B}_\Psi^{w*} \mathbf{x}_\Psi + (\mathcal{D}_\Psi^*)^{-1} \mathcal{D}_\Psi^{zw*} \tilde{w} + (\mathcal{D}_\Psi^*)^{-1} \underline{z} \\ &= (\hat{\mathcal{B}}_\Psi^*)^* \mathbf{x}_\Psi + (\hat{\mathcal{D}}_\Psi^{z\tilde{w}})^* \tilde{w} + (\hat{\mathcal{D}}_\Psi^{w\tilde{w}})^* \underline{z},\end{aligned}$$

Substituting this expression into the first block row of (23) yields

$$\begin{aligned}\mathcal{T}_\Psi^* \dot{\mathbf{x}}_\Psi &= (\mathcal{A}_\Psi^* - \mathcal{C}_\Psi^{w*} (\mathcal{D}_\Psi^*)^{-1} \mathcal{B}_\Psi^{w*}) \mathbf{x}_\Psi \\ &\quad + (\mathcal{C}_\Psi^{z*} - \mathcal{C}_\Psi^{w*} (\mathcal{D}_\Psi^*)^{-1} \mathcal{D}_\Psi^{zw*}) \tilde{w} - \mathcal{C}_\Psi^{w*} (\mathcal{D}_\Psi^*)^{-1} \underline{z} \\ &= (\hat{\mathcal{A}}_\Psi)^* \mathbf{x}_\Psi + (\hat{\mathcal{C}}_\Psi^z)^* \tilde{w} + (\hat{\mathcal{C}}_\Psi^w)^* \underline{z}.\end{aligned}$$

Likewise, substituting into the third block row gives

$$\begin{aligned}\underline{w} &= (\mathcal{B}_\Psi^{z*} - \mathcal{D}_\Psi^{wz*} (\mathcal{D}_\Psi^*)^{-1} \mathcal{B}_\Psi^{w*}) \mathbf{x}_\Psi \\ &\quad + (\mathcal{D}_\Psi^{z*} - \mathcal{D}_\Psi^{wz*} (\mathcal{D}_\Psi^*)^{-1} \mathcal{D}_\Psi^{zw*}) \tilde{w} - \mathcal{D}_\Psi^{wz*} (\mathcal{D}_\Psi^*)^{-1} \underline{z} \\ &= (\hat{\mathcal{B}}_\Psi^z)^* \mathbf{x}_\Psi + (\hat{\mathcal{D}}_\Psi^z)^* \tilde{w} + (\hat{\mathcal{D}}_\Psi^{wz})^* \underline{z}.\end{aligned}$$

Collecting these three state-space relations gives

$$\begin{bmatrix} \mathcal{T}_\Psi^* \dot{\mathbf{x}}_\Psi(t) \\ \tilde{z}(t) \\ \underline{w}(t) \end{bmatrix} = \begin{bmatrix} (\hat{\mathcal{A}}_\Psi)^* & (\hat{\mathcal{C}}_\Psi^z)^* & (\hat{\mathcal{C}}_\Psi^w)^* \\ (\hat{\mathcal{B}}_\Psi^z)^* & (\hat{\mathcal{D}}_\Psi^{z\tilde{w}})^* & (\hat{\mathcal{D}}_\Psi^{w\tilde{w}})^* \\ (\hat{\mathcal{B}}_\Psi^z)^* & (\hat{\mathcal{D}}_\Psi^z)^* & (\hat{\mathcal{D}}_\Psi^{wz})^* \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Psi(t) \\ \tilde{w}(t) \\ \underline{z}(t) \end{bmatrix}. \quad (35)$$

We can now augment the dual system with (35). Substitution of $\underline{z} = \mathcal{B}^* \mathbf{x} + \mathcal{D}^* \underline{w}$ into the last block row of (35) gives the algebraic loop

$$\underline{w} = (\hat{\mathcal{D}}_\Psi^{wz})^* \mathcal{B}^* \mathbf{x} + (\hat{\mathcal{B}}_\Psi^z)^* \mathbf{x}_\Psi + (\hat{\mathcal{D}}_\Psi^z)^* \tilde{w} + (\hat{\mathcal{D}}_\Psi^{wz})^* \mathcal{D}^* \underline{w}.$$

Assume this loop is well-posed. Since $\Lambda_z^* = (I - (\hat{\mathcal{D}}_\Psi^{wz})^* \mathcal{D}^*)^{-1}$, solving for $\underline{w}$ gives

$$\underline{w} = \Lambda_z^* (\hat{\mathcal{D}}_\Psi^{wz})^* \mathcal{B}^* \mathbf{x} + \Lambda_z^* (\hat{\mathcal{B}}_\Psi^z)^* \mathbf{x}_\Psi + \Lambda_z^* (\hat{\mathcal{D}}_\Psi^z)^* \tilde{w}.$$

Using $\mathcal{D}^* \Lambda_z^* = \Lambda_p^* \mathcal{D}^*$ and $I + \mathcal{D}^* \Lambda_z^* (\hat{\mathcal{D}}_\Psi^{wz})^* = \Lambda_p^*$, we also have

$$\underline{z} = \Lambda_p^* \mathcal{B}^* \mathbf{x} + \Lambda_p^* \mathcal{D}^* (\hat{\mathcal{B}}_\Psi^z)^* \mathbf{x}_\Psi + \Lambda_p^* \mathcal{D}^* (\hat{\mathcal{D}}_\Psi^z)^* \tilde{w}.$$

To prove sufficiency, suppose the first statement holds. By Lemma 1, the realization in (26) satisfies $\|P_\tau \tilde{z}\| \leq \|P_\tau \tilde{w}\|$ for all $\tau \geq 0$. Applying the argument of Theorem 3 with $\gamma = 1$ to (26) gives its dual realization $(\mathcal{D} \Lambda_p = \Lambda_z \mathcal{D}, \Lambda_p \hat{\mathcal{D}}_\Psi^{wz} = \hat{\mathcal{D}}_\Psi^{wz} \Lambda_z)$

$$\begin{bmatrix} \mathcal{T}_\Psi^* \dot{\mathbf{x}}_\Psi(t) \\ \tilde{z}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* + \mathcal{C}^* \Lambda_z^* (\hat{\mathcal{D}}_\Psi^{wz})^* \mathcal{B}^* & \mathcal{C}^* \Lambda_z^* (\hat{\mathcal{B}}_\Psi^z)^* & \mathcal{C}^* \Lambda_z^* (\hat{\mathcal{D}}_\Psi^z)^* \\ (\hat{\mathcal{C}}_\Psi^w)^* \Lambda_p^* \mathcal{B}^* & (\hat{\mathcal{A}}_\Psi)^* + (\hat{\mathcal{C}}_\Psi^w)^* \Lambda_p^* \mathcal{D}^* (\hat{\mathcal{B}}_\Psi^z)^* & (\hat{\mathcal{C}}_\Psi^z)^* + (\hat{\mathcal{C}}_\Psi^w)^* \Lambda_p^* \mathcal{D}^* (\hat{\mathcal{D}}_\Psi^z)^* \\ (\hat{\mathcal{D}}_\Psi^{w\tilde{w}})^* \Lambda_p^* \mathcal{B}^* & (\hat{\mathcal{B}}_\Psi^z)^* + (\hat{\mathcal{D}}_\Psi^{w\tilde{w}})^* \Lambda_p^* \mathcal{D}^* (\hat{\mathcal{B}}_\Psi^z)^* & (\hat{\mathcal{D}}_\Psi^{z\tilde{w}})^* + (\hat{\mathcal{D}}_\Psi^{w\tilde{w}})^* \Lambda_p^* \mathcal{D}^* (\hat{\mathcal{D}}_\Psi^z)^* \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{x}_\Psi(t) \\ \tilde{w}(t) \end{bmatrix}. \quad (36)$$

satisfies $\|P_\tau \tilde{z}\| \leq \|P_\tau \tilde{w}\|$ for all $\tau \geq 0$. By Lemma 1. The realization in (36) is the **Dual System** augmented with the Potapov-Ginzburg transform (35) of the **Dual Scaling**. Hence by Lemma 1 this is equivalent to the **Scaled Dual System** satisfying the same inequality. Since $(G^\sharp)^\sharp = G$ and $\mathbf{D}(\mathbf{D}(\Psi)) = \Psi$ we have that sufficiency implies necessity. $\blacksquare$

Example 1 (Dual multiplier for the zero-padded LTI-parametric IQC). *Consider an LTI-parametric IQC with causal bounded linear tall factor $\psi = \begin{bmatrix} I \\ \psi_t \end{bmatrix}$ and pointwise multiplier with $\mathcal{R} = -\mathcal{Q}$,*

$$\mathcal{P} := \begin{bmatrix} \mathcal{Q} & \mathcal{S} \\ \mathcal{S}^* & -\mathcal{Q} \end{bmatrix}, \quad \mathcal{Q} \succ 0, \quad \mathcal{S} = -\mathcal{S}^*. \quad (37)$$

Let $\mathcal{M} \in \mathcal{L}(\mathbf{L}_{[a,b]})$ be the multiplication operator induced by $\mathcal{P}$, so, for $v \in \mathbf{L}_{[a,b]}$,

$$(\mathcal{M}v)(t) := \mathcal{P}v(t). \quad (38)$$

Define the original tall filtered map

$$\Psi_{\text{tall}} := \begin{bmatrix} \psi & 0 \\ 0 & \psi \end{bmatrix}. \quad (39)$$

In this example we use the system-side convention this is the opposite sign from the uncertainty-side IQC convention in Definition 7. Assume the following inequality holds for all $T \geq 0$ and all $z(t), w(t) \in \mathbb{R}L_2$ where $z(t)$ satisfies Definition 15.,

$$\left\langle p_T \Psi_{\text{tall}} \begin{bmatrix} z \\ w \end{bmatrix}, p_T \mathcal{M} \Psi_{\text{tall}} \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq 0. \quad (40)$$

Since Definition 11 applies to square invertible scalings, introduce

$$\mathcal{N} := \begin{bmatrix} I & 0 \end{bmatrix}, \quad \mathcal{N}^* = \begin{bmatrix} I \\ 0 \end{bmatrix}, \quad \psi = \psi_{\square} \mathcal{N}^*, \quad (41)$$

$$\psi_{\square} := \begin{bmatrix} I & 0 \\ \psi_t & I \end{bmatrix} \begin{bmatrix} I & \psi_t^{\#} \\ 0 & I \end{bmatrix} = \begin{bmatrix} I & \psi_t^{\#} \\ \psi_t & I + \psi_t \psi_t^{\#} \end{bmatrix}. \quad (42)$$

By construction $\psi_t^{\#}$ is causal bounded. Then $\psi_{\square}$ is square and invertible, with causal bounded inverse

$$\psi_{\square}^{-1} = \begin{bmatrix} I & -\psi_t^{\#} \\ 0 & I \end{bmatrix} \begin{bmatrix} I & 0 \\ -\psi_t & I \end{bmatrix} = \begin{bmatrix} I + \psi_t^{\#} \psi_t & -\psi_t^{\#} \\ -\psi_t & I \end{bmatrix}. \quad (43)$$

Square up the plant by setting $z_{\square} = \mathcal{N}^* z$ and $w = \mathcal{N} w_{\square}$:

$$\begin{bmatrix} \mathcal{T} \dot{\mathbf{x}}(t) \\ z_{\square}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \mathcal{N} \\ \mathcal{N}^* \mathcal{C} & \mathcal{N}^* \mathcal{D} \mathcal{N} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ w_{\square}(t) \end{bmatrix}. \quad (44)$$

The original plant is recovered on the embedded square subspace by taking $w_{\square} = \mathcal{N}^* w = \begin{bmatrix} w \\ 0 \end{bmatrix}$. Hence, for $z_{\square} = \mathcal{N}^* z = \begin{bmatrix} z \\ 0 \end{bmatrix}$

and $w_{\square} = \mathcal{N}^* w = \begin{bmatrix} w \\ 0 \end{bmatrix}$, the squared-up primal J-contraction inequality is

$$\Psi_{\square} := \begin{bmatrix} \psi_{\square} & 0 \\ 0 & \psi_{\square} \end{bmatrix}, \quad \left\langle p_T \Psi_{\square} \begin{bmatrix} z_{\square} \\ w_{\square} \end{bmatrix}, p_T \mathcal{M} \Psi_{\square} \begin{bmatrix} z_{\square} \\ w_{\square} \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq 0. \quad (45)$$

Factor the pointwise multiplier as $\mathcal{P} = \mathcal{V}^* J \mathcal{V}$ with

$$\mathcal{V} := \begin{bmatrix} \mathcal{V}_{11} & \mathcal{V}_{12} \\ 0 & \mathcal{V}_{22} \end{bmatrix}, \quad J := \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix}, \quad (46)$$

where $\mathcal{V}_{11}$ and $\mathcal{V}_{22}$ are invertible. Define the J-factorization

$$\Phi := \mathcal{V} \Psi_{\square} = \begin{bmatrix} \mathcal{V}_{11} \psi_{\square} & \mathcal{V}_{12} \psi_{\square} \\ 0 & \mathcal{V}_{22} \psi_{\square} \end{bmatrix}. \quad (47)$$

Using Definition 11 gives the dual scaling

$$\mathbf{D}(\Phi) = \mathbf{D}(\mathcal{V}) \mathbf{D} \left(\begin{bmatrix} \psi_{\square} & 0 \\ 0 & \psi_{\square} \end{bmatrix} \right) = \underbrace{\begin{bmatrix} \mathcal{V}_{22}^{-*} & \mathcal{V}_{22}^{-*} \mathcal{V}_{12}^* \mathcal{V}_{11}^{-*} \\ 0 & \mathcal{V}_{11}^{-*} \end{bmatrix}}_{\mathcal{V}_d} \underbrace{\begin{bmatrix} \psi_{\square}^{-\#} & 0 \\ 0 & \psi_{\square}^{-\#} \end{bmatrix}}_{\Psi_{d,\square}}. \quad (48)$$

Hence the pointwise multiplication operator $\mathcal{M}_d \in \mathcal{L}(\mathbf{L}_{[a,b]})$ by

$$\mathcal{P}_d := \mathcal{V}_d^* J \mathcal{V}_d, \quad (\mathcal{M}_d v)(t) := \mathcal{P}_d v(t). \quad (49)$$

Thus the squared-up dual contraction inequality is

$$\left\langle p_T \Psi_{d,\square} \begin{bmatrix} z_{\square} \\ w_{\square} \end{bmatrix}, p_T \mathcal{M}_d \Psi_{d,\square} \begin{bmatrix} z_{\square} \\ w_{\square} \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq 0. \quad (50)$$

Equivalently,

$$\mathcal{P}_d = \begin{bmatrix} \mathcal{Q}_d & \mathcal{S}_d \\ \mathcal{S}_d^* & \mathcal{R}_d \end{bmatrix} = \mathcal{K}^{-1}, \quad \mathcal{K} := \begin{bmatrix} \mathcal{Q} & \mathcal{S}^* \\ \mathcal{S} & -\mathcal{Q} \end{bmatrix}. \quad (51)$$

With

$$\Theta := \mathcal{Q} + \mathcal{S}^* \mathcal{Q}^{-1} \mathcal{S} = \mathcal{Q} - \mathcal{S} \mathcal{Q}^{-1} \mathcal{S}, \quad (52)$$

the block inverse is

$$\mathcal{P}_d = \begin{bmatrix} \Theta^{-1} & \Theta^{-1} \mathcal{S}^* \mathcal{Q}^{-1} \\ \mathcal{Q}^{-1} \mathcal{S} \Theta^{-1} & -\Theta^{-1} \end{bmatrix}, \quad (53)$$

so

$$\mathcal{Q}_d = \Theta^{-1} \succ 0, \quad \mathcal{S}_d = \Theta^{-1} \mathcal{S}^* \mathcal{Q}^{-1} = -\Theta^{-1} \mathcal{S} \mathcal{Q}^{-1}, \quad \mathcal{R}_d = -\Theta^{-1} = -\mathcal{Q}_d \prec 0. \quad (54)$$

Moreover, since

$$SQ^{-1}\Theta = \Theta Q^{-1}S = S - SQ^{-1}SQ^{-1}S, \quad (55)$$

we have $\Theta^{-1}SQ^{-1} = Q^{-1}S\Theta^{-1}$, and therefore

$$S_d^* = Q^{-1}S\Theta^{-1} = \Theta^{-1}SQ^{-1} = -S_d, \quad \text{equivalently} \quad S_d = -S_d^*. \quad (56)$$

To express the dual condition on the original signal dimension, restrict the squared-up dual variables to the subspace $\underline{z}_\square = \mathcal{N}^* \underline{z}$ and $\underline{w}_\square = \mathcal{N}^* \underline{w}$, so that $\underline{z} = \mathcal{N} \underline{z}_\square$ and $\underline{w} = \mathcal{N} \underline{w}_\square$. This gives

$$\psi_d := \psi_\square^\# \mathcal{N}^* = \begin{bmatrix} I + \psi_t^\# \psi_t \\ -\psi_t \end{bmatrix}. \quad (57)$$

Thus

$$\Psi_d = \begin{bmatrix} \psi_d & 0 \\ 0 & \psi_d \end{bmatrix}. \quad (58)$$

The induced dual contraction inequality is therefore, for all $T \geq 0$ and all $\underline{z}(t), \underline{w}(t) \in \mathbb{R}L_2$, where $\underline{z}(t)$ satisfies Definition 21,

$$\left\langle p_T \Psi_d \begin{bmatrix} \underline{z} \\ \underline{w} \end{bmatrix}, p_T \mathcal{M}_d \Psi_d \begin{bmatrix} \underline{z} \\ \underline{w} \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq 0. \quad (59)$$

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